Parte 5

Resolução de Sistemas Lineares - 3

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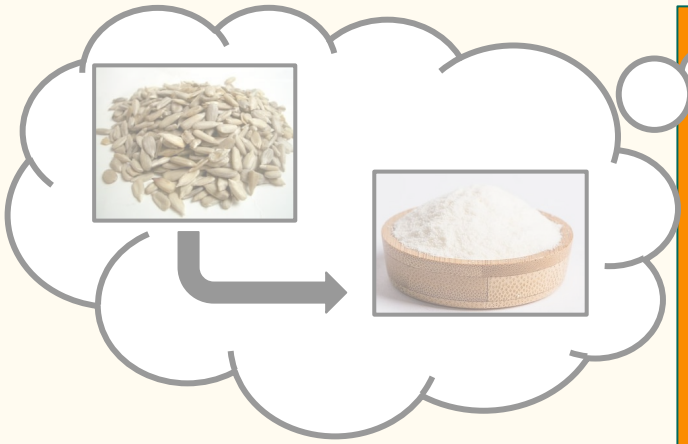
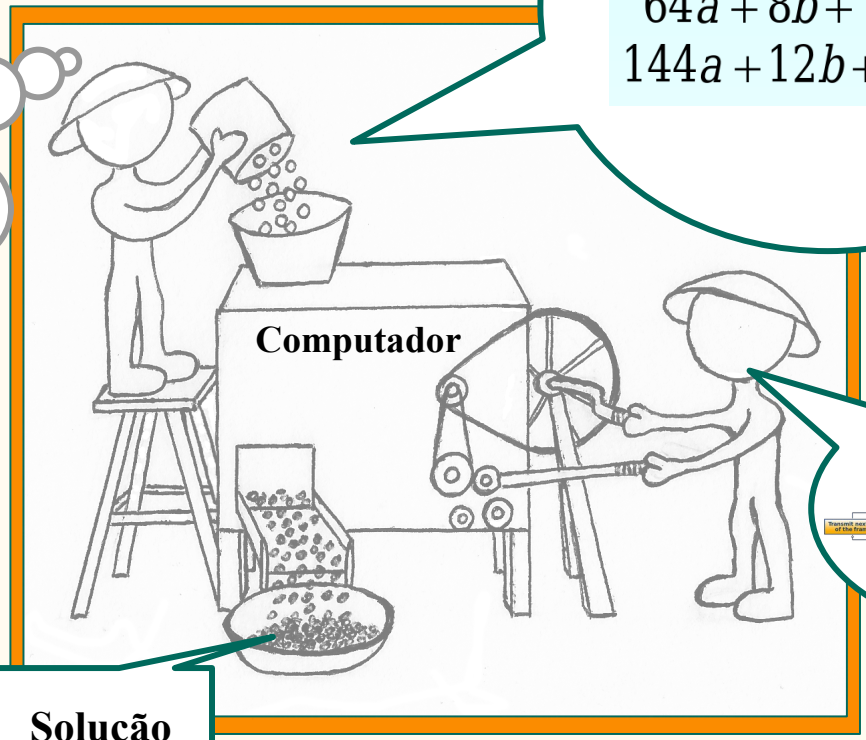
CI1164 - Introdução à Computação Científica
Profs. Armando Delgado e Guilherme Derenievicz
Departamento de Informática - UFPR

Visão geral da disciplina:

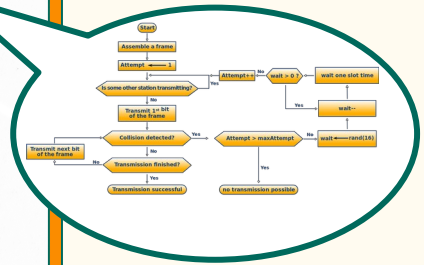


Descrição do Problema



$$25a + 5b + c = 106$$
$$64a + 8b + c = 177$$
$$144a + 12b + c = 600$$

Sistemas Lineares



Método Numérico

Solução

Resolução de Sistemas Lineares

$$
A = \begin{bmatrix}
-\alpha & 0 & 0 & 1 & \alpha & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\alpha & 0 & -1 & 0 & -\alpha & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\alpha & -1 & 0 & 0 & \alpha & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \alpha & 0 & 1 & 0 & \alpha & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -\alpha & 0 & 0 & 1 & \alpha & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\alpha & 0 & -1 & 0 & -\alpha & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & \alpha & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -\alpha & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\alpha & -1 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \alpha & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\alpha & -1
\end{bmatrix}
$$

Resolução de Sistemas Lineares

$$A = \begin{bmatrix}
-\alpha & 0 & 0 & 1 & \alpha & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\alpha & 0 & -1 & 0 & -\alpha & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
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0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\alpha & -1 & 0 & 0 & \alpha & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \alpha & 0 & 1 & 0 & \alpha & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -\alpha & 0 & 0 & 1 & \alpha & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\alpha & 0 & -1 & 0 & -\alpha & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & \alpha & 0 \\
 & & & & & & & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -\alpha & 0 \\
 & & & & & & & 0 & 0 & 0 & 0 & 0 & -\alpha & -1 & 0 & 0 & 1 \\
 & & & & & & & 0 & 0 & 0 & 0 & 0 & \alpha & 0 & 1 & 0 & 0 \\
 & & & & & & & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\alpha & -1
\end{bmatrix}$$

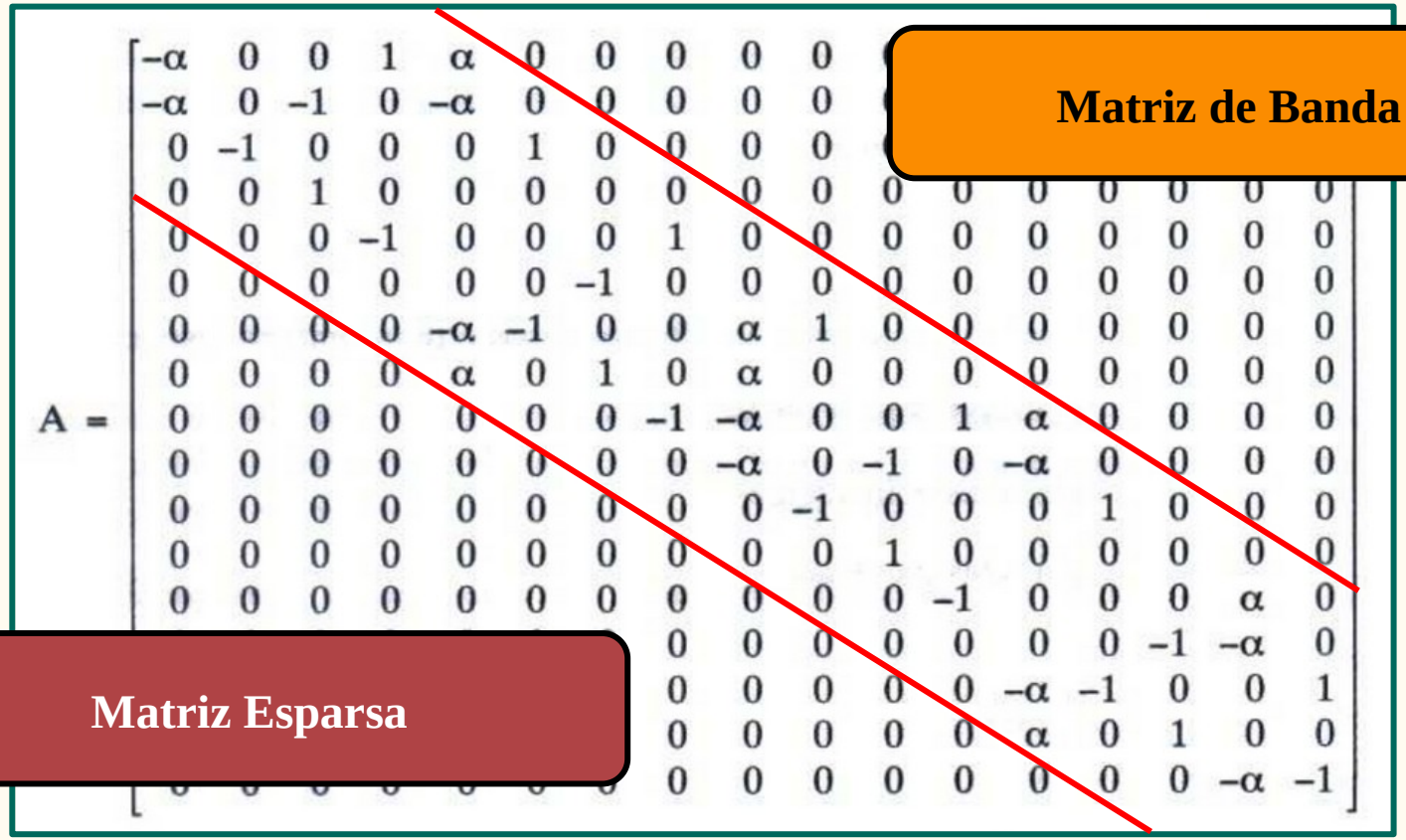
Matriz Esparsa

Resolução de Sistemas Lineares

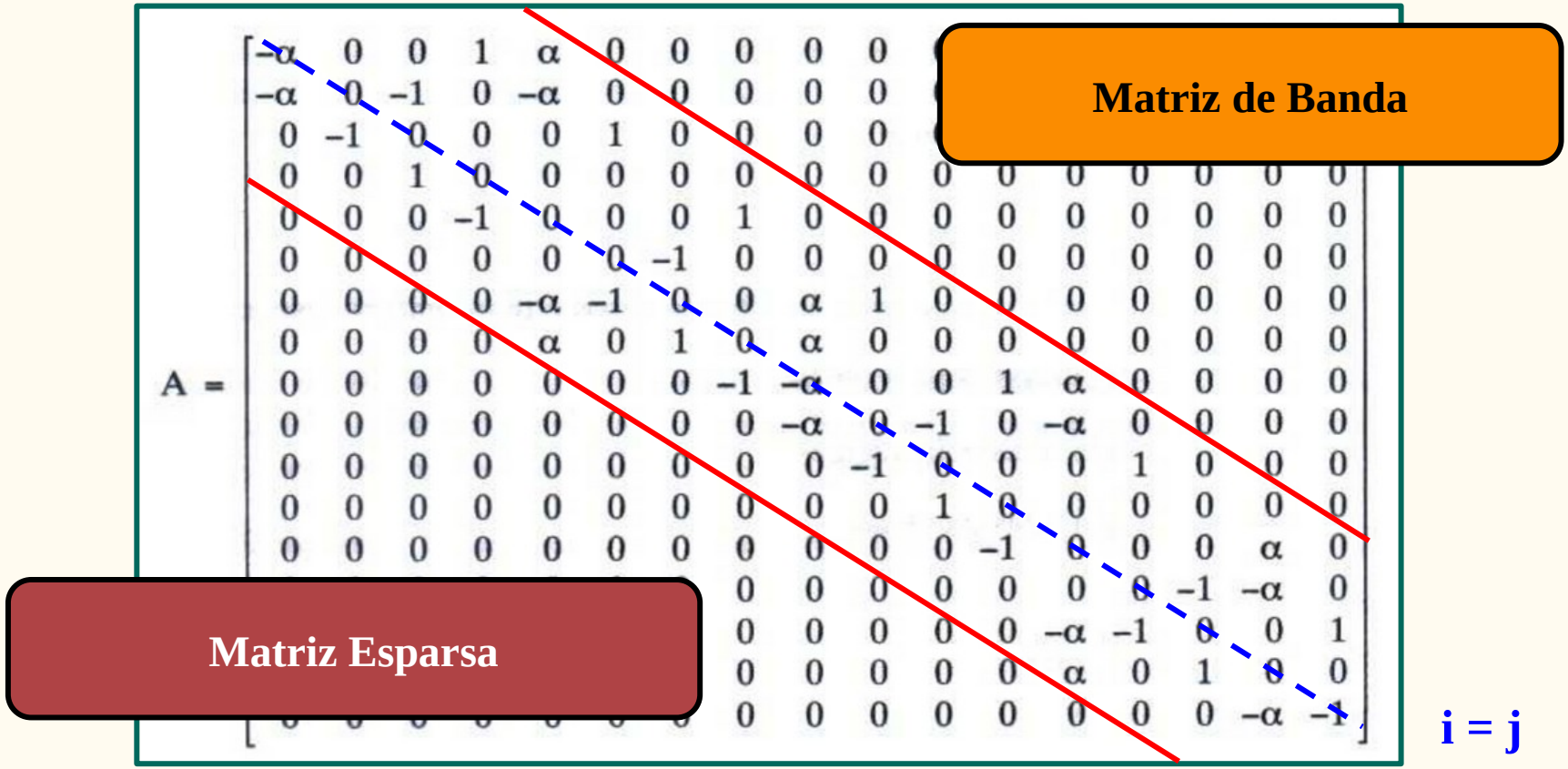
$$A = \begin{bmatrix}
-\alpha & 0 & 0 & 1 & \alpha & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\alpha & 0 & -1 & 0 & -\alpha & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\alpha & -1 & 0 & 0 & \alpha & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \alpha & 0 & 1 & 0 & \alpha & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -\alpha & 0 & 0 & 1 & \alpha & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\alpha & 0 & -1 & 0 & -\alpha & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & \alpha & 0 \\
 & & & & & & & & & & & & 0 & 0 & 0 & 0 & 0 \\
 & & & & & & & & & & & & 0 & 0 & 0 & 0 & 0 \\
 & & & & & & & & & & & & 0 & 0 & 0 & 0 & 0 \\
 & & & & & & & & & & & & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

Matriz Esparsa

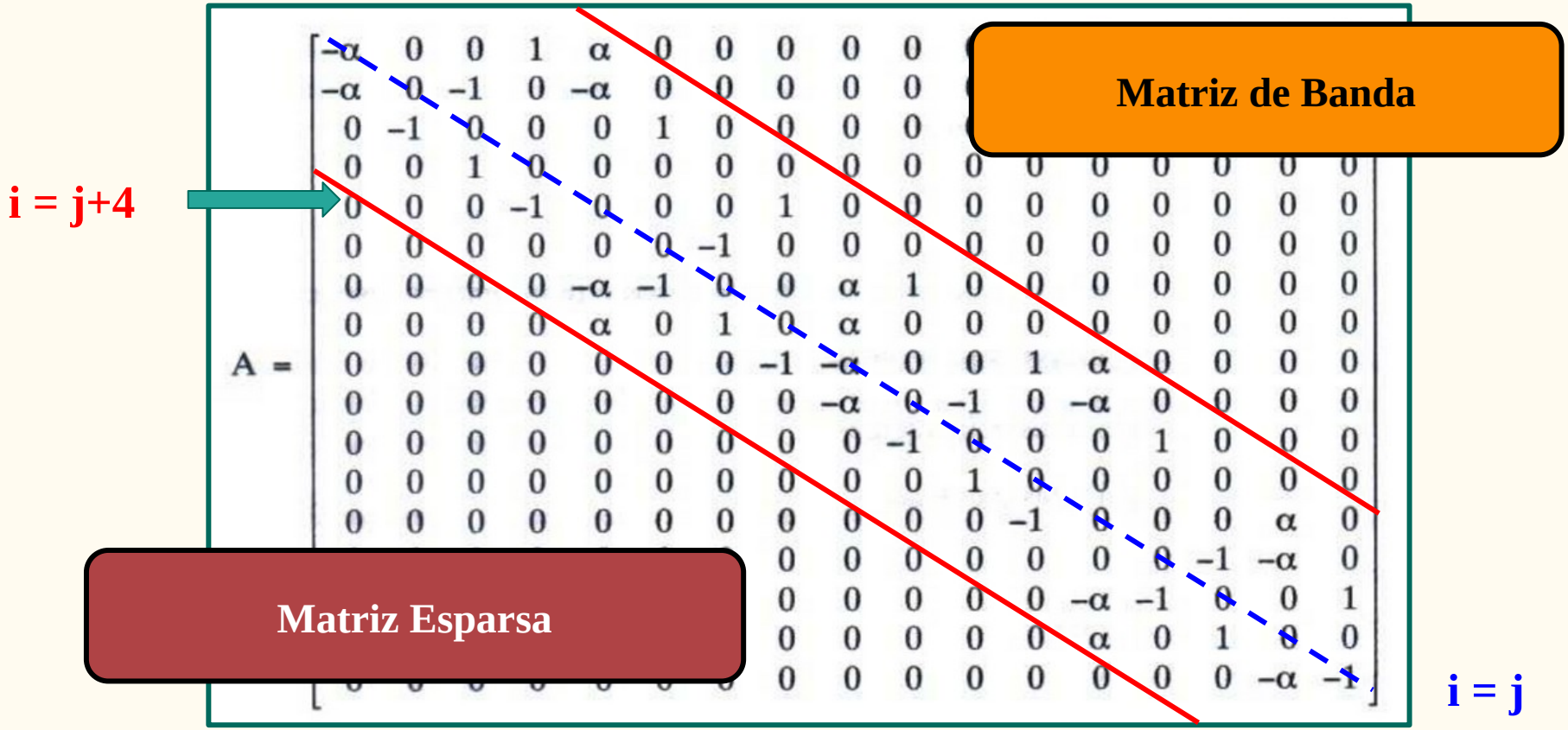
Matrizes k-Diagonais



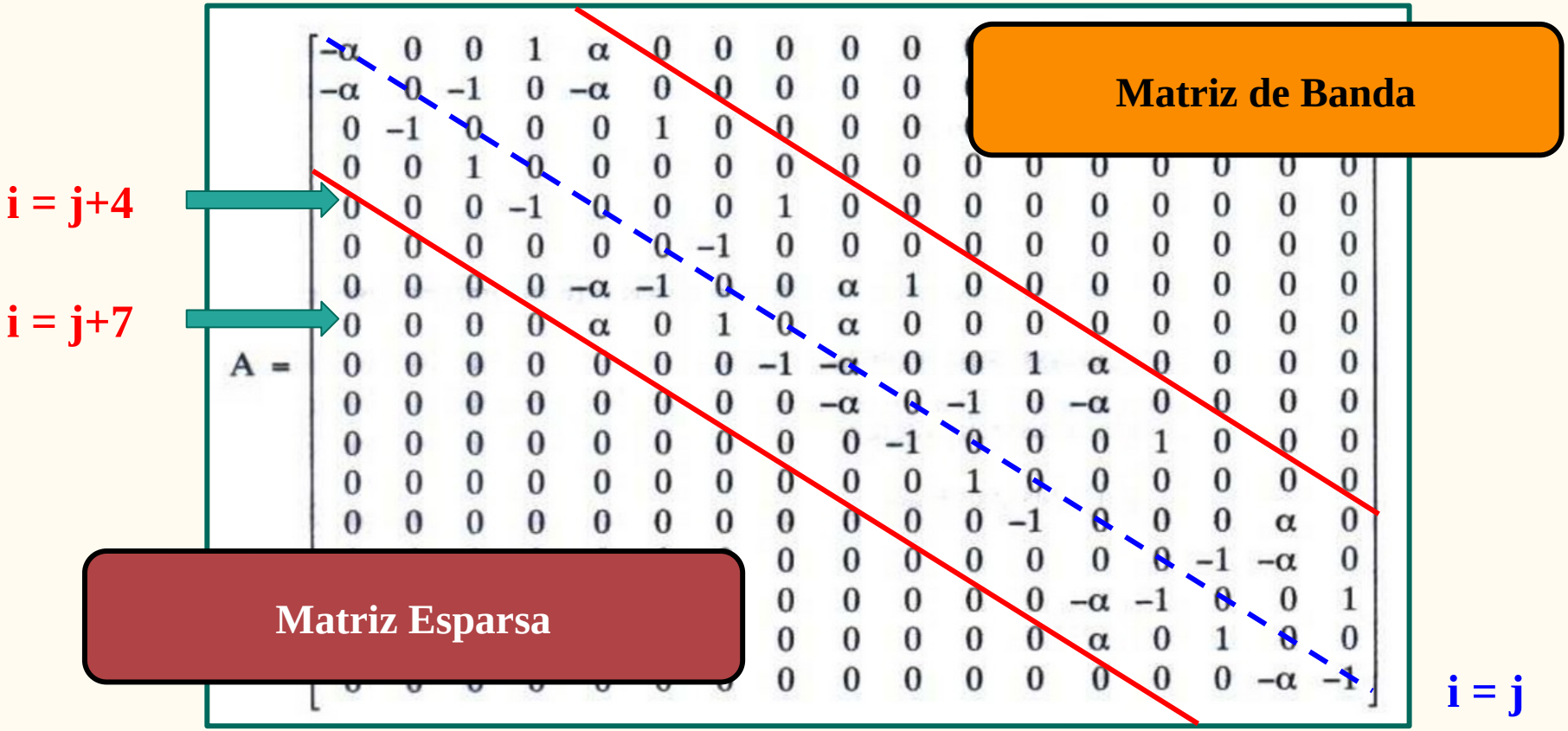
Matrizes k-Diagonais



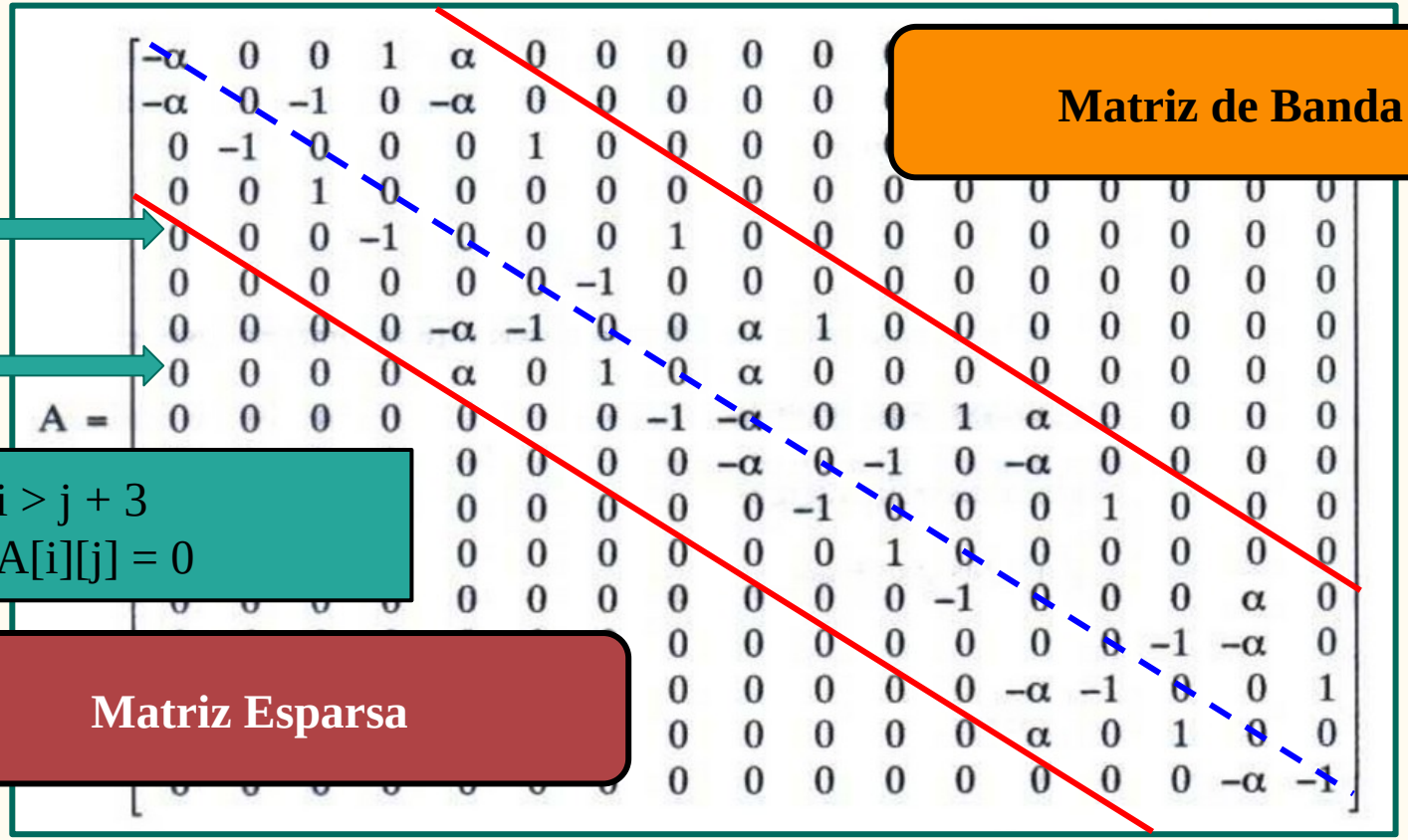
Matrizes k-Diagonais



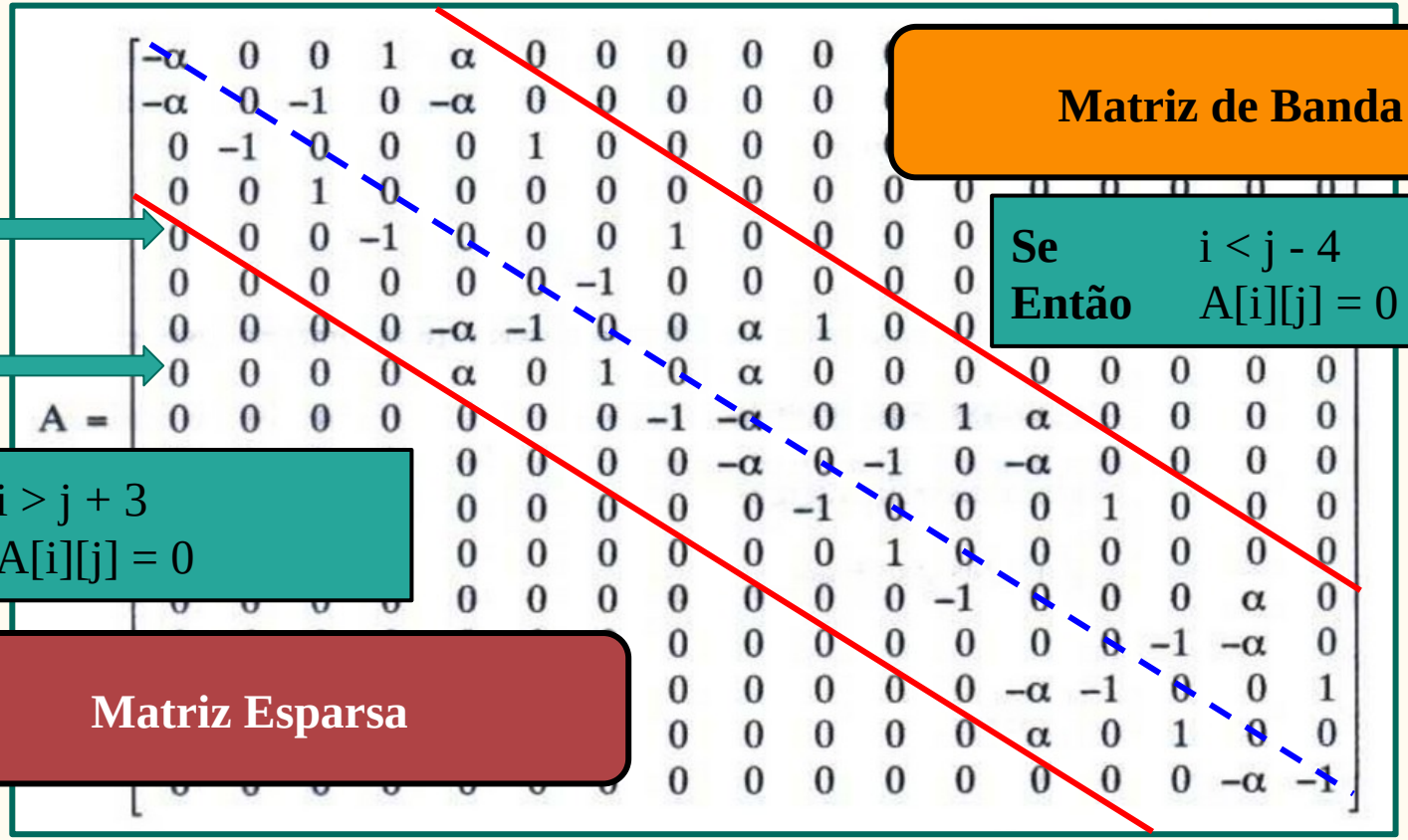
Matrizes k-Diagonais



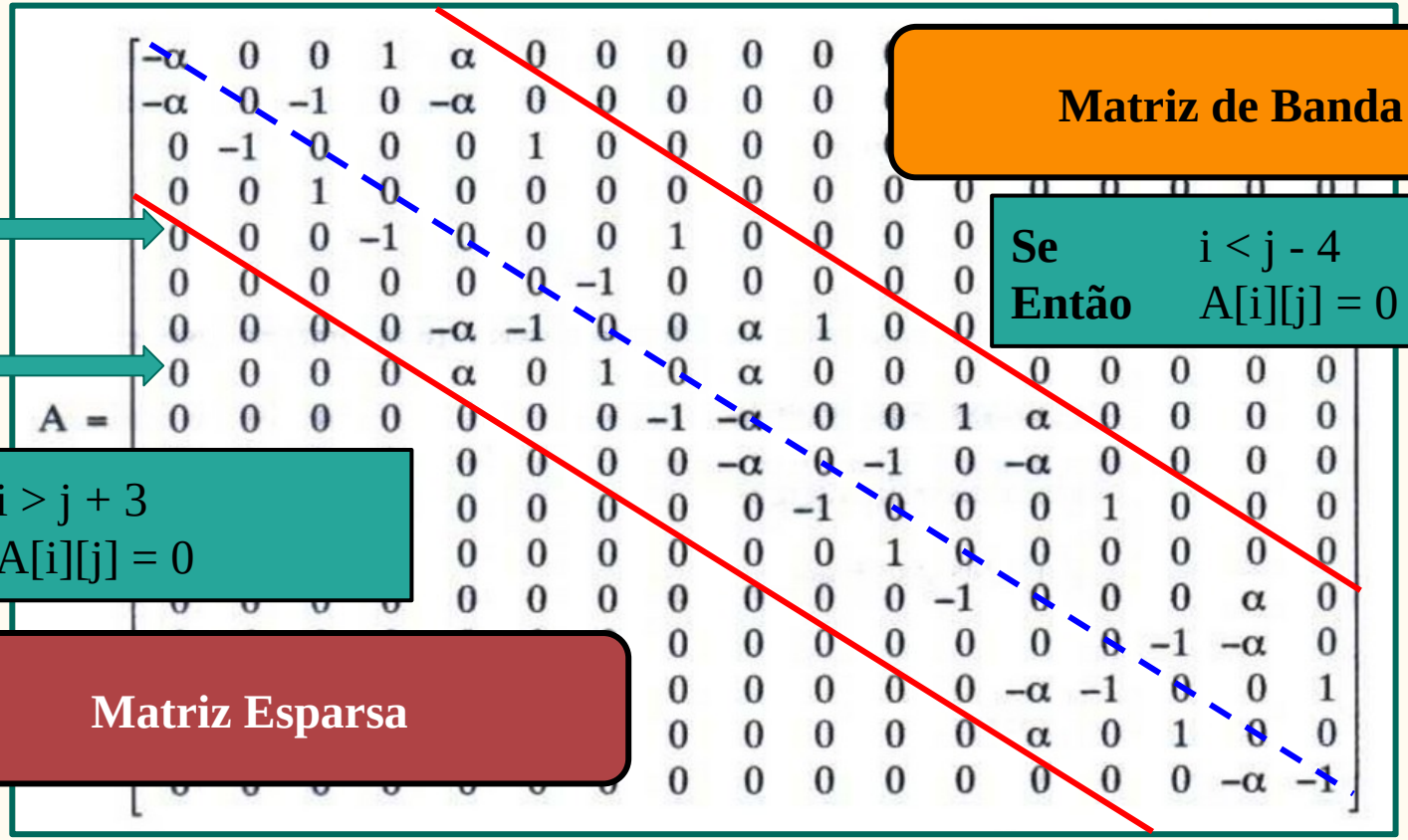
Matrizes k-Diagonais



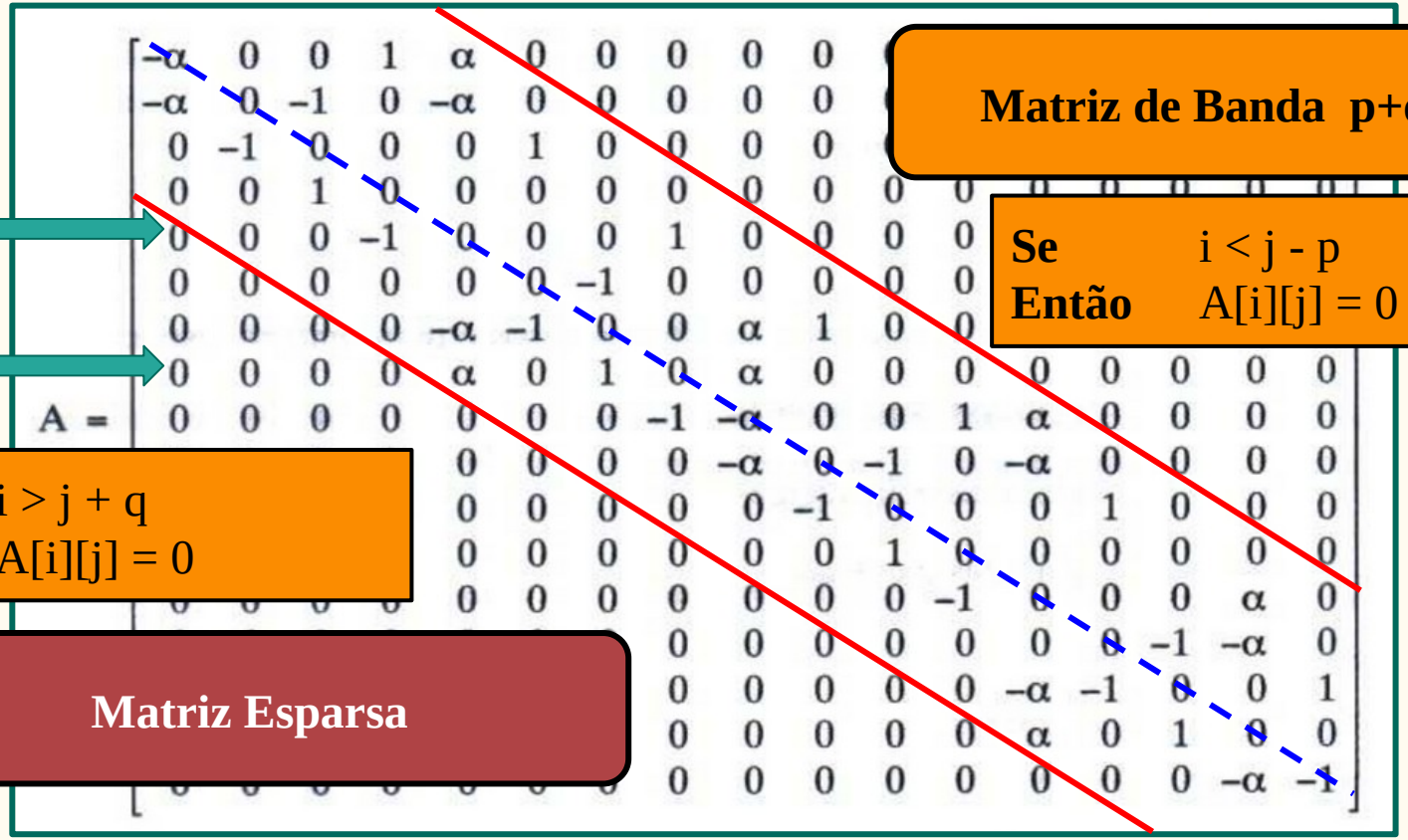
Matrizes k-Diagonais



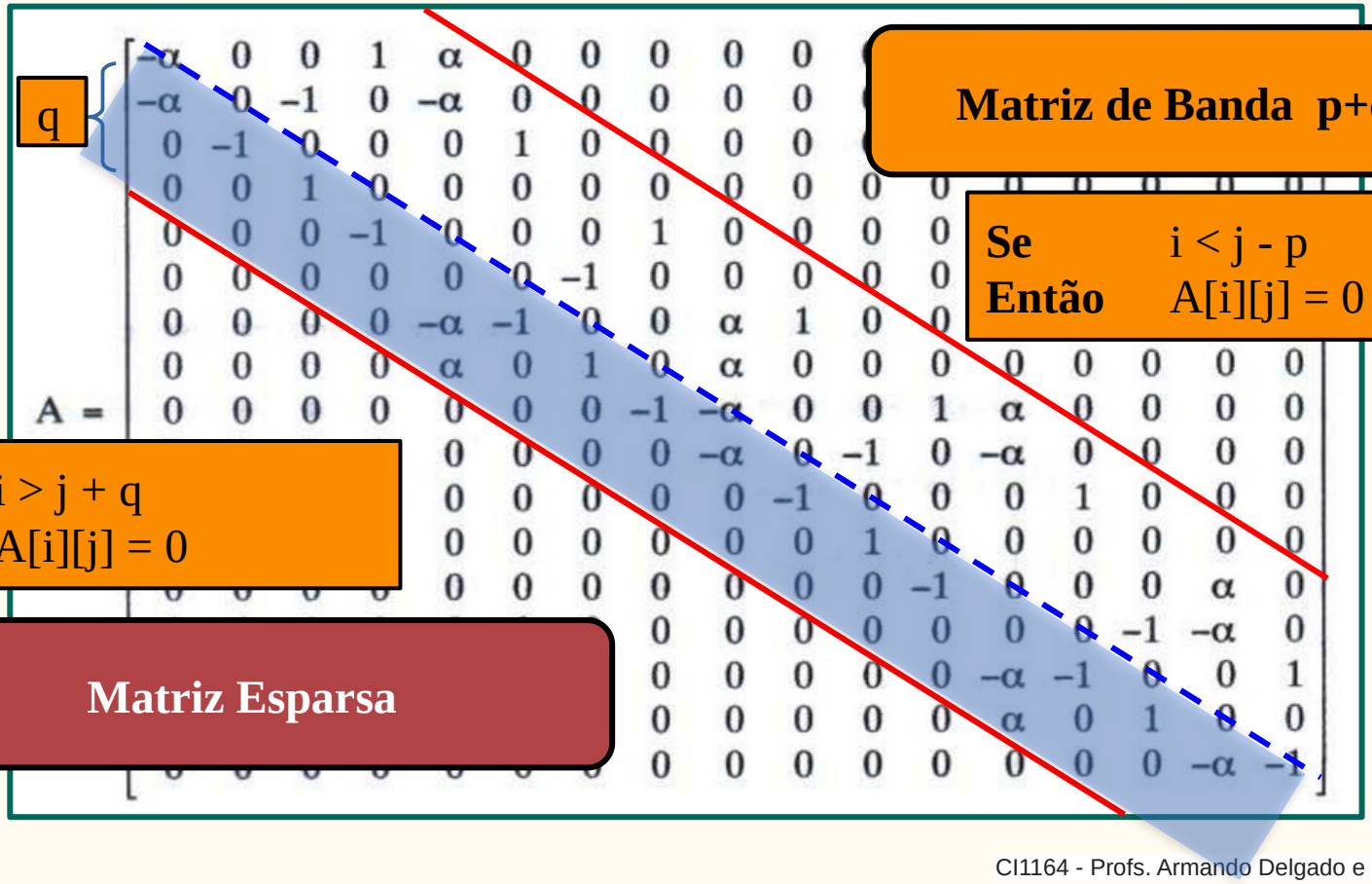
Matrizes k-Diagonais



Matrizes k-Diagonais



Matrizes k-Diagonais



Matriz de Banda p+q+1

Se i < j - p
Então A[i][j] = 0

Se i > j + q
Então A[i][j] = 0

Matriz Esparsa

Simplificação nos métodos

- Armazenamento dos elementos da matriz em memória
  - ➔ Basta as diagonais

- Implementação dos métodos ficam mais simples
  - ➔ Menos laços aninhados
  - ➔ Menos operações em ponto flutuante

Matrizes Tridiagonais (p = q = 1)

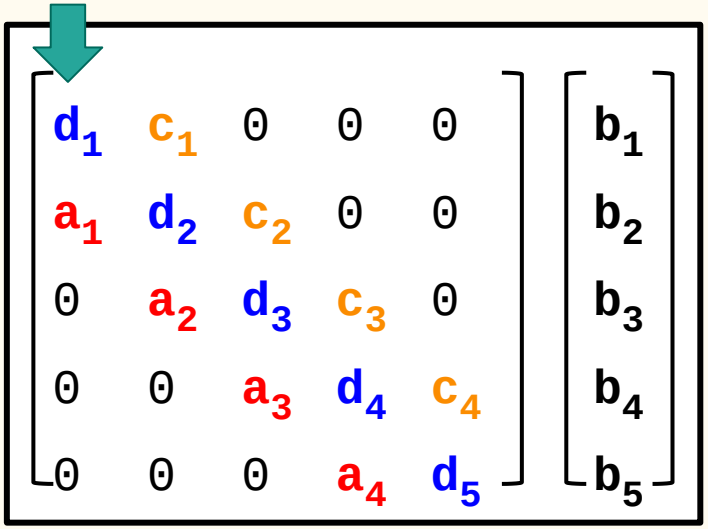
$$\begin{bmatrix} d_1 & c_1 & 0 & 0 & 0 \\ a_1 & d_2 & c_2 & 0 & 0 \\ 0 & a_2 & d_3 & c_3 & 0 \\ 0 & 0 & a_3 & d_4 & c_4 \\ 0 & 0 & 0 & a_4 & d_5 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}$$

Matrizes Tridiagonais (p = q = 1)

$$\begin{bmatrix} d_1 & c_1 & 0 & 0 & 0 \\ a_1 & d_2 & c_2 & 0 & 0 \\ 0 & a_2 & d_3 & c_3 & 0 \\ 0 & 0 & a_3 & d_4 & c_4 \\ 0 & 0 & 0 & a_4 & d_5 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}$$

```
/* Seja um S.L. de ordem 'n'
*/
void eliminacaoGauss( double **A, double *b, u
   /* para cada linha a partir da primeira */
   for (int i=0; i < n; ++i) {
      for(int k=i+1; k < n; ++k) {
         double m = A[k][i] / A[i][i];
         A[k][i] = 0.0;
         for(int j=i+1; j < n; ++j)
            A[k][j] -= A[i][j] * m;
         b[k] -= b[i] * m;
      }
   }
}
```

Matrizes Tridiagonais (p = q = 1)

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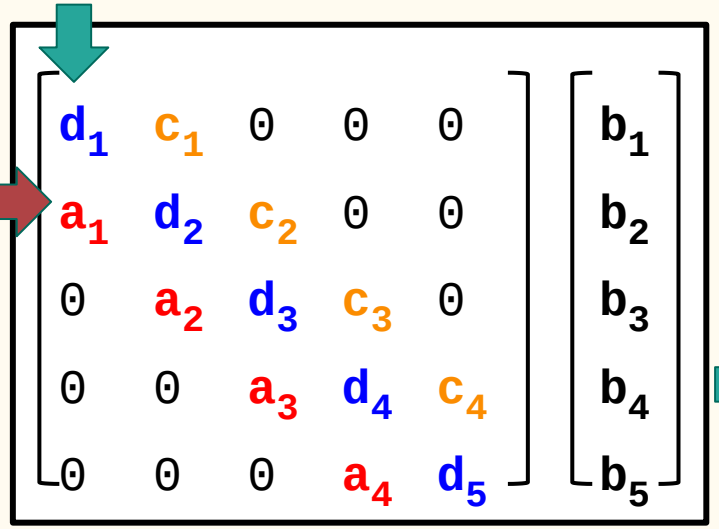
         for(int j=i+1; j < n; ++j)
            A[k][j] -= A[i][j] * m;
         b[k] -= b[i] * m;
      }
   }
}
```

Matrizes Tridiagonais (p = q = 1)

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         for(int j=i+1; j < n; ++j)
            A[k][j] -= A[i][j] * m;
         b[k] -= b[i] * m;
      }
   }
}
```
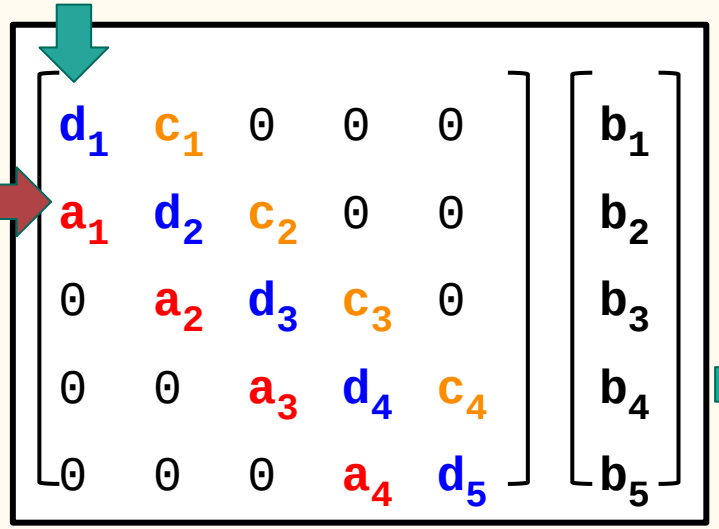

Matrizes Tridiagonais (p = q = 1)




```
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   for (int i=0; i < n; ++i) {
      for(int k=i+1; k < n; ++k) {
         double m = A[k][i] / A[i][i];
         A[k][i] = 0.0;
         for(int j=i+1; j < n; ++j)
            A[k][j] -= A[i][j] * m;
         b[k] -= b[i] * m;
      }
   }
}
```
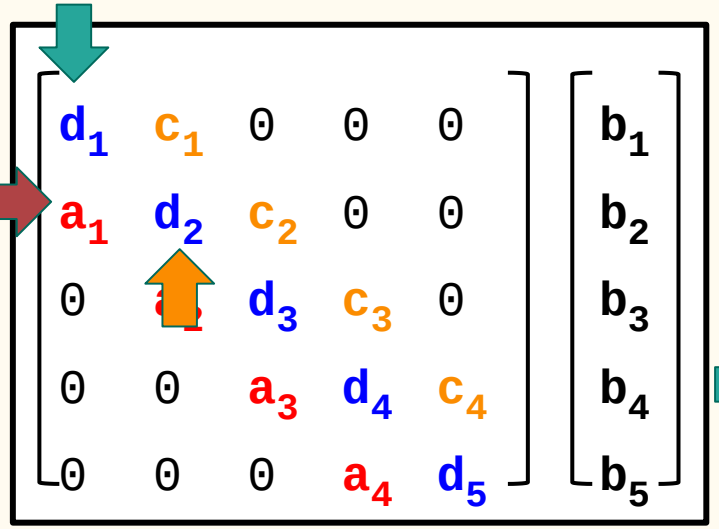

k = i+1

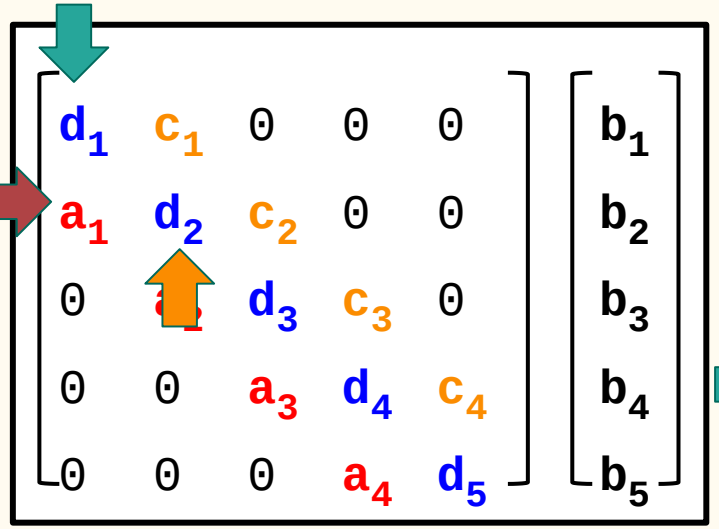
Matrizes Tridiagonais (p = q = 1)




```
/* Seja um S.L. de ordem 'n'
*/
void eliminacaoGauss( double **A, double *b, u
   /* para cada linha a partir da primeira */
   for (int i=0; i < n; ++i) {
      for(int k=i+1; k < n; ++k) {
         double m = A[k][i] / A[i][i];
         A[k][i] = 0.0;
         for(int j=i+1; j < n; ++j)
            A[k][j] -= A[i][j] * m;
         b[k] -= b[i] * m;
      }
   }
}
```

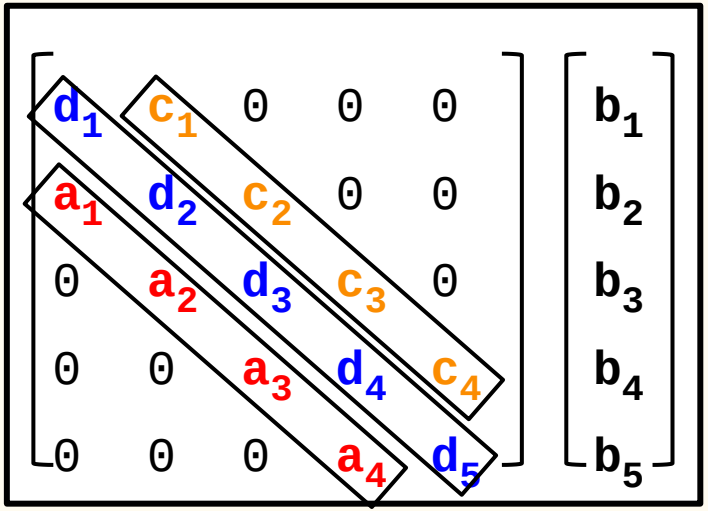
k = i+1

Matrizes Tridiagonais (p = q = 1)




```
/* Seja um S.L. de ordem 'n'
*/
void eliminacaoGauss( double **A, double *b, u
   /* para cada linha a partir da primeira */
   for (int i=0; i < n; ++i) {
      for(int k=i+1; k < n; ++k) {        k = i+1
         double m = A[k][i] / A[i][i];
         A[k][i] = 0.0;
         for(int j=i+1; j < n; ++j)       j = i+1
            A[k][j] -= A[i][j] * m;
         b[k] -= b[i] * m;
      }
   }
}
```

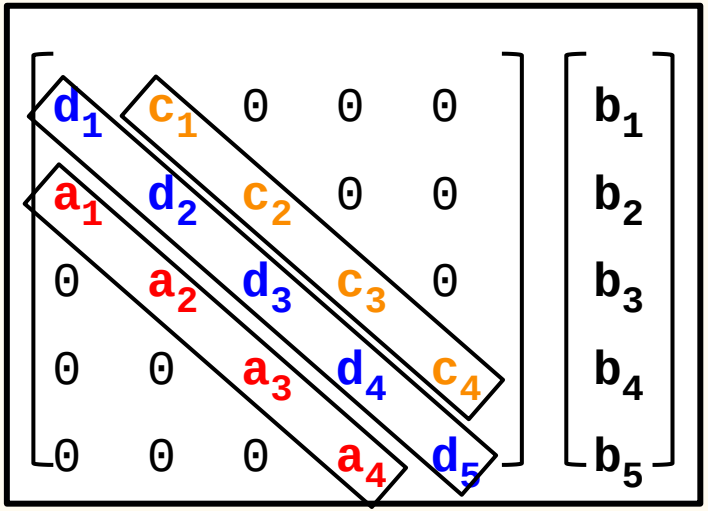
Matrizes Tridiagonais (p = q = 1)




```
double b[5];
double d[5];
double a[4];
double c[4];
```

```
/* Seja um S.L. de ordem 'n'
*/
void eliminacaoGauss( double **A, double *b, u
   /* para cada linha a partir da primeira */
   for (int i=0; i < n; ++i) {
      for(int k=i+1; k < n; ++k) {        k = i+1
         double m = A[k][i] / A[i][i];
         A[k][i] = 0.0;
         for(int j=i+1; j < n; ++j)       j = i+1
            A[k][j] -= A[i][j] * m;
         b[k] -= b[i] * m;
      }
   }
}
```

Matrizes Tridiagonais (p = q = 1)

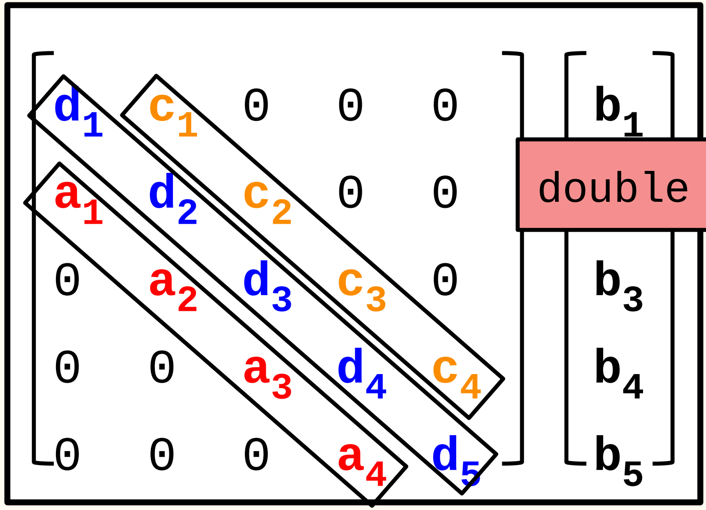



```
double b[5];
double d[5];
double a[4];
double c[4];
```

```
double *d, double *a, double *c
```

```
/* Seja um S.L. de ordem
*/
void eliminacaoGauss( double **A, double *b, u
   /* para cada linha a partir da primeira */
   for (int i=0; i < n; ++i) {
      for(int k=i+1; k < n; ++k) {
         double m = A[k][i] / A[i][i];
         A[k][i] = 0.0;
         for(int j=i+1; j < n; ++j)
            A[k][j] -= A[i][j] * m;
         b[k] -= b[i] * m;
      }
   }
}
```

Matrizes Tridiagonais (p = q = 1)



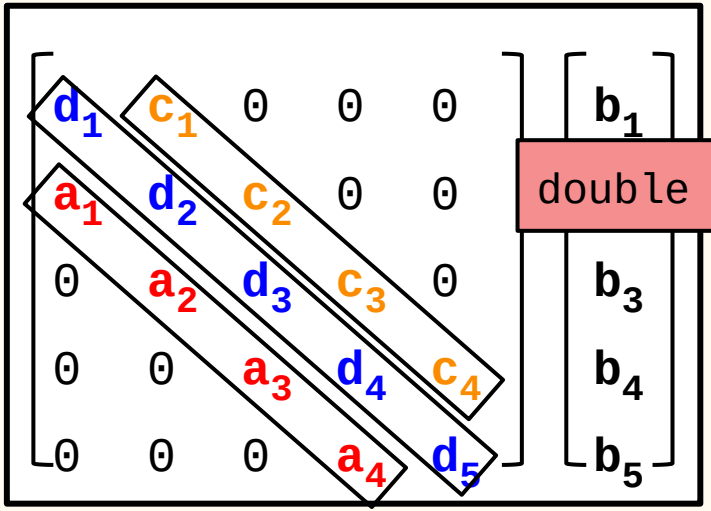

```
double b[5];
double d[5];
double a[4];
double c[4];
```

`double *d, double *a, double *c`

`double m = a[i] / d[i];`

```
ordem
void elim        Gauss( double **A, double *b, u
   /* para c     linha a partir da primeira */
   for (int i=0  i < n; ++i) {
```
~~for(int k=i+1; k < n; ++k) {~~
```
         double m = A[k][i] / A[i][i];
         A[k][i] = 0.0;
```
~~for(int j=i+1; j < n; ++j)~~
```
            A[k][j] -= A[i][j] * m;
         b[k] -= b[i] * m;
      }
   }
}
```

Matrizes Tridiagonais (p = q = 1)




```
double b[5];
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```
double *d, double *a, double *c
```

```
double m = a[i] / d[i];
```

```
a[i] = 0.0;
```

```
ordem
void elim    Gauss( double **A, double *b, u
   /* para c   lin    a primeira */
   for (int i=0    i
      for(int k=i+1;    < n; ++k) {
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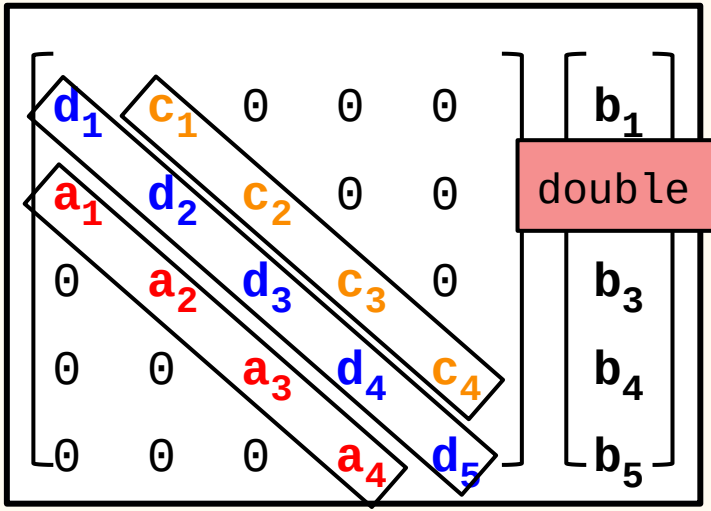
Matrizes Tridiagonais (p = q = 1)

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```
double *d, double *a, double *c
```

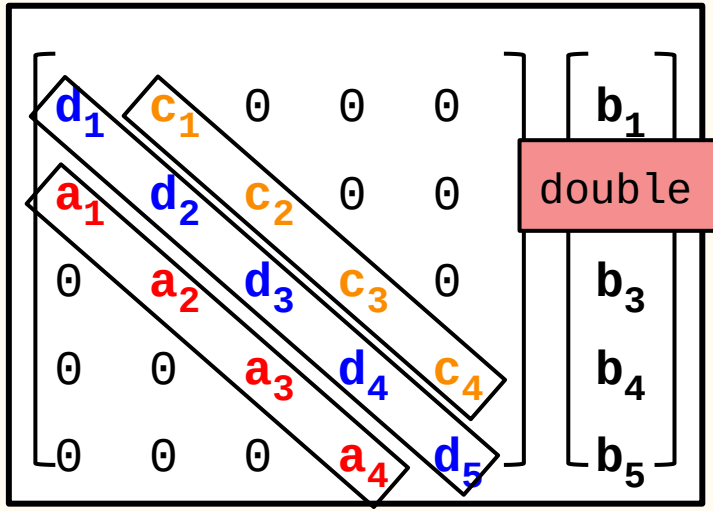
```
double m = a[i] / d[i];
```

```
a[i] = 0.0;
```

```
d[i+1] -= c[i]*m;
```

```
ordem
void elim    Gauss( double **A, double *b, u
   /* para c   lin    a primeira */
   for (int i=0    i
      for(int k=i+1;    < n; ++k) {
         double m = A[k][i] / A[i][i];
         A[k][i] = 0.0;
         for(int j=i+1; j < n; ++j)
            A[k][j] -= A[i][j] * m;
         b[k] -= b[i] * m;
      }
   }
}
```

Matrizes Tridiagonais (p = q = 1)




```
double b[5];
double d[5];
double a[4];
double c[4];
```

```
double *d, double *a, double *c

double m = a[i] / d[i];

ordem
void elim  Gauss( double **A, double *b, u
   /* para c  lin  a primeira */
   for (int i=0  i
      for(int k=i+1;  < n; ++k) {
         double m = A[k][i] / A[i][i];
         A[k][i] = 0.0;
         for(int j=i+1; j < n; ++j)
            A[k][j] -= A[i][j] * m;
         b[k] -= b[i] * m;
      }
   }
}

a[i] = 0.0;

b[i+1] -= b[i]*m;

d[i+1] -= c[i]*m;
```

SL Tridiagonais ➔ Eliminação de Gauss

$$\begin{bmatrix} d_0 & c_0 & 0 & 0 & 0 \\ a_0 & d_1 & c_1 & 0 & 0 \\ 0 & a_1 & d_2 & c_2 & 0 \\ 0 & 0 & a_2 & d_3 & c_3 \\ 0 & 0 & 0 & a_3 & d_4 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

```
double b[5];
double d[5];
double a[4];
double c[4];
```

```
void eliminacaoGauss(double *d, double *a,
                     double *c, double *b, double *x,
                     uint n)
{
   // TRIANGULARIZAÇÃO:  5(n-1) operações
   for (int i=0; i < n-1; ++i) {
      double m = a[ i ] / d[ i ];
      a[ i ] = 0.0;
      d[ i+1 ] -= c[ i ] * m;
      b[ i+1 ] -= b[ i ] * m;
   }
   // RETROSUBSTITUIÇÃO: ≈ 3n operações (n grande)
   x[ n-1 ] = b[ n-1 ] / d[ n-1 ];
   for (int i=n-2; i >= 0; --i)
      x[ i ] = (b[ i ] - c[ i ] * x[ i+1 ]) / d[ i ];
}
```

SL Tridiagonais ➔ Gauss-Seidel

$$\begin{bmatrix} d_0 & c_0 & 0 & 0 & 0 \\ a_0 & d_1 & c_1 & 0 & 0 \\ 0 & a_1 & d_2 & c_2 & 0 \\ 0 & 0 & a_2 & d_3 & c_3 \\ 0 & 0 & 0 & a_3 & d_4 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

```
double b[5];
double d[5];
double a[4];
double c[4];
```

```
void gaussSeidel (double **A,
                  double *b, double *x, uint n, double tol)
{
  double erro = 1.0 + tol;
  int j, s;
  while (erro < tol) {  // n(2n+1) ≈ 2n² FLOPs / iteração
    for (int i=0; i < n; ++i) {
      for (s=0, j=0; j < n; ++j)
        if (i != j)     s += A[ i ][ j ] * x[ j ];

      x[ i ] = (b[ i ] – s) / A[ i ][ i ];

      // Calcula erro == norma máxima de ( x(k) – x(k-1) )
       .......
    }
  }
```

SL Tridiagonais ➔ Gauss-Seidel

$$\begin{bmatrix} d_0 & c_0 & 0 & 0 & 0 \\ a_0 & d_1 & c_1 & 0 & 0 \\ 0 & a_1 & d_2 & c_2 & 0 \\ 0 & 0 & a_2 & d_3 & c_3 \\ 0 & 0 & 0 & a_3 & d_4 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

```
double b[5];
double d[5];
double a[4];
double c[4];
```

```
void gaussSeidel (double *d, double *a, double *c,
                  double *b, double *x, uint n, double tol)
{
  double erro = 1.0 + tol;
   while (erro < tol) {   // 5(n-2)+6 ≈ 5n FLOPs / iteração
     x[ 0 ] = (b[ 0 ] – c[ 0 ] * x[ 1 ]) / d[ 0 ];

     for (int i=1; i < n-1; ++i)
       x[ i ] = (b[ i ] – a[ i-1 ] * x[ i-1] – c[ i ] * x[ i+1 ]) / d[ i ];

     x[ n-1 ] = (b[ n-1 ] – a[ n-2 ] * x[ n-2 ] ) / d[ n-1 ];

       // Calcula erro == norma máxima de ( x(k) – x(k-1) )
        .......
   }
```